DOE Title Page

Project Title:
FLAIME (Fast Learning AI for Mineral Efficiency): Autonomous Discovery and Qualification of Critical-Mineral-Lean Alloys for Clean Energy

Lead Applicant/Institution: Brigham Young University (BYU)
Street Address: 701 E University Parkway, 350 EB, Provo, UT 84602
Postal Address: Same as street address
Lead PI: Sterling G. Baird, Ph.D. | (801) 422-5344 | sterling.baird@byu.edu
Administrative POC: BYU research administration
RFA Number: DE-FOA-0003612
Challenge and Focus Area: 3-B Securing America's Critical Minerals Supply | AI-Enabled Materials Discovery and Engineering

Senior/Key Personnel

Institution	Senior/Key Personnel
BYU (Lead)	S. G. Baird (PI)
Valimet (Funded Subrecipient)	Valimet's CTO; Valimet's technical manager — gas atomization

Summary Budget Information

Institution	Year 1 Budget ($)
BYU (Lead, including the Valimet subaward)	743,999.99
Total federal request	**743,999.99**

1. Background and Introduction

Overview.

The purpose of this proposal is to demonstrate how FLAIME (Fast Learning AI for Mineral Efficiency) targets DOE Topic 3-B's critical mineral demand reduction objective by closing the discovery → manufacturing → qualification loop for **Sc-free, Mn-Cr-Zr-lean aluminum alloys for laser powder bed fusion (LPBF)**.[1] Manganese and zirconium are 100% import-reliant and chromium 74% (USGS 2024 [1]). Scalmalloy is Sc-dominated, and the four confirmed commercial Additive Manufacturing (AM) Al alloys (AlSi10Mg, Scalmalloy, AL250C, AlSi7Mg) [2–6] collectively concentrate U.S. AM production risk on these elements, making it susceptible to a single supply chain disruption [7]. We propose FLAIME, a Level 3 self-driving laboratory (SDL)[2] that will discover LPBF aluminum alloys meeting structural performance targets while minimizing critical mineral content. Related SDL/BO efforts include APEX [9], (DED, no LPBF/tensile) A-Lab (oxides, not structural metals) [10], AMASE (thin-film phase maps) [11], high-throughput DED active-learning without LPBF/tensile qualification [12]; BIRDSHOT (arc-melted refractories, pre-selected system) [13], ICME-anchored closed-loop Al for AM [14], and BO-guided CALPHAD within a single expert-selected Al system [15]. However, none yet *unify powder production, LPBF printing, and tensile testing under a single AI decision loop with joint composition–process optimization*. By closing this decision loop via a complete powder-to-part pipeline, FLAIME targets a measurable AI acceleration factor of $\geq 3\times$ over conventional alloy development for the decision making portion utilizing Bayesian optimization. These estimates are consistent with >3–$5\times$ speedups reported for closed-loop materials campaigns[13, 16] and compounded with closed-loop execution speedup from hardware and software automation for an estimated $\sim 10\times$ total reduction in discovery time [8].

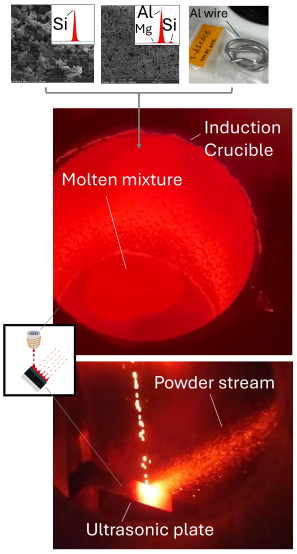


Figure 1: FLAIME atomization process: feedstock (Si + AlSi10Mg + 1100 Al wire, byu-vcl) → induction-crucible melt → ultrasonic-plate breakup (AMAZEMET rePOWDER).

Project relevance and context. The DOE Critical Materials Strategy (Pillars 2, 3, 5) [7] and alloy level criticality metrics [17, 18] motivate composition level supply risk optimization as a first-class Bayesian Optimization over Function Network (BOFN) objective alongside strength and printability. AMMTO's CMI Hub precedent (Al–Ce to TRL 9 [19]) motivates the same urgency for lean critical mineral LPBF alternatives. All confirmed commercial AM Al deployments (AlSi10Mg, Scalmalloy, AL250C, AlSi7Mg) use LPBF exclusively [2–6], validating FLAIME's primary focus on LPBF.

FLAIME bridges computation and physical qualification using BOFN minimizing mineral burden objective [20, 21], building on supply risk aware optimization [22]. Recent cost-aware BO advances make AI guided powder-to-part discovery tractable [23].

Several preliminary data have been collected specifically for this proposal to demonstrate feasibility and de-risk the proposed approach: powder characterization, mixing, and atomization of a custom blend, Thermo-Calc TCAl10/AM Module CALPHAD simulations, LPBF printing, SEM/EDS characterization, and tensile tests for commercial AlSi10Mg powder, pKGFN [24] optimization benchmarking campaigns using AM-like synthetic network functions running on BYU's HPC. (Fig. 3).

[1] Buy America Act compliance: this project does not involve construction, alteration, maintenance, or repair of public infrastructure (FOA §III.F).

[2] Level 3 in the taxonomy of Tom et al. [8]: a closed-loop, AI-guided platform in which software selects experiments autonomously while a small set of defined human handoffs execute physical transfers.

The Lead PI brings direct SDL experience since 2022 (incl. three years of Acceleration Consortium leadership) and lead authored the structural materials section in a comprehensive SDL review [8], a frugal twin framework [25], and a critical raw materials article [17]. Prior high dimensional BO work [26, 27] informs FLAIME's joint acquisition whose compositional subspace implicitly spans 22 803 element combination subsystems. **Valimet brings significant expertise in gas atomization** and scaled up industrial atomization processes.

2. Project Objectives

Goal. Discover novel, high performing, critical mineral lean LPBF Al alloys for structural applications and demonstrate a measurable **AI advantage** through established metrics [8]—specifically, AI acceleration factor, discovery yield, and mineral-burden reduction— advancing the dual Topic 3B mandate of AI enabled materials discovery *and* reduced demand for critical minerals.

O1. Demonstrate a closed-loop BOFN/pKGFN platform for AI guided, cost-aware alloy exploration across the full powder-to-part pipeline.

O2. Cover a joint search space spanning 22 803 element combinations (Al + 15 alloying element upper bounds) and evaluate 100–200 compositions via CALPHAD partial evaluations under a mineral burden objective.

O3. Validate 100–200 compositions experimentally through the minimum viable specimen (MVS) pipeline over a 9-month Phase I.

O4. Achieve discovery yield $\geq$3% *and* satisfy the dual baseline Gate G3 mineral burden constraints vs. Scalmalloy and AlSi10Mg.

O5. Demonstrate AI acceleration factor [8] $\geq$3$\times$ vs. a pre-registered random baseline [28], with bootstrap CIs across $\geq$20 seeds.

Per-family YS/EL thresholds, the Tier 1 pass-rate complementary metric, and throughput derivation are given in and App. 6.

Collectively, fulfillment of these objectives demonstrate that a software orchestrated, AI guided closed-loop platform can simultaneously accelerate alloy discovery *and* reduce critical mineral dependence—the dual mandate of Topic 3B. Phase I focuses on Al alloys processed by LPBF. Al with the 15-element palette (Mn, Cr, Zr, Mg, Si, Cu, Ti, Fe, Ni, Ce, Sc, Li, Er, Zn, Sn) span 0% to 100% USGS import reliance, ensuring the mineral burden objective produces genuine supply risk tradeoffs.

Technology transfer and commercialization. All FLAIME software releases open source (permissive licensing) via BoTorch tutorials [29] per Meta's letter of collaboration and Honegumi [30]. Discovered alloys will be published openly rather than licensed: the most broadly adopted AM Al alloy (AlSi10Mg) is unpatented, and even patented high-performers (e.g., Scalmalloy) reached broad adoption primarily through standardization and industry partnerships. The team has prepared an extensive **multi-site validation network of collaborating universities, national labs, and companies** that have agreed to accept a batch of gas-atomized powder for one or more FLAIME alloy compositions, print a specimen on their metal 3D printer (LPBF/DED/Binder Jet), and return a printed sample with associated build metadata so that alloy performance can be cross-validated across printers and process conditions rather than localized to a single machine (Fig. 2); each partner site contributes in-kind printer/operator time with potential for further collaborations in a Phase II continuation. The team will also submit several presentations to SME's RAPID + TCT 2027 (largest North-American AM conference, April 2027) and pursue a complimentary academic booth showcasing FLAIME software, tutorials, and validated compositions—just outside the Phase I period of performance which ends 31 March 2027 and therefore not charged to this award.

3. Proposed Research and Methods

Activity 1: AI-Guided Computational Exploration (*PI, BYU ∼$220K*).

The FLAIME optimizer is a BOFN directed acyclic graph (DAG; Fig. 2) [13, 31] covering Thermo-Calc AM Module simulation, powder, LPBF, and mechanical response nodes. The *partial* Knowledge Gradient for Function Networks (pKGFN) [24, 32] is the acquisition function that selects which node and inputs to evaluate next in a cost-aware manner: "partial" because not every node is evaluated at each iteration (e.g., a Thermo-Calc AM Module query can be run without triggering an expensive tensile test), "knowledge gradient" because the decision model maximizes the expected improvement in the global objective per unit cost ranging from Thermo-Calc AM Module simulation to tensile testing, and "network" because the DAG structure—where each node's output feeds downstream nodes—is exploited to propagate information efficiently (Fig. 3). A mineral burden node (analytical, noise-free) exposes two cost-weighted scalars to act as hard supply chain metrics: $\mathrm{MBS}_{\mathrm{lin}} = \sum_{i \neq \mathrm{Al}} \mathrm{NIR}_i (c_i / c_{\mathrm{Al}}) x_i$ as a gate (G3a) on Sc-elimination and $\mathrm{CMDRI}_{\log} = \sum_{i \neq \mathrm{Al}} \mathrm{NIR}_i \log_{10}(1 + c_i / c_{\mathrm{Al}}) x_i$ for continuous BOFN objective and a gate (G3b) for backfill ceiling vs. AlSi10Mg. USGS MCS 2024 NIR [1] and 2022–2024 prices frozen at month 1 will be used. Graedel [33], Helbig [22], ESSENZ [34], and $\mathrm{HHI}_{\mathrm{prod/res}}$ enter as digital-thread reporting metadata/contextual BOFN features instead of hard thresholds per the multi-axis framework of Baird & Sparks [17]. Preliminary pKGFN benchmarking campaigns on BYU's HPC and preliminary CALPHAD and AM Module evaluations have been performed.

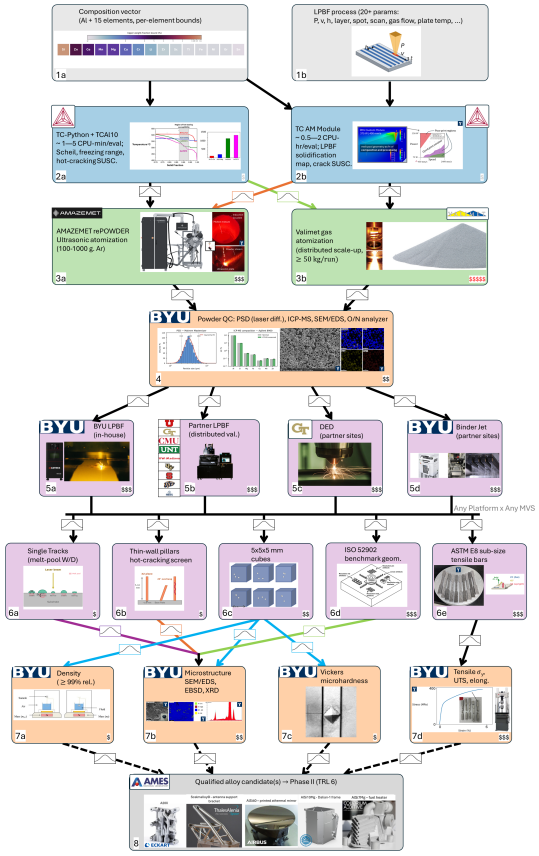


Figure 2: FLAIME's Bayesian Optimization Function Network (BOFN) for AM alloy discovery. See narrative prose for in-text references to specific nodes.

Joint Acquisition with GA Inner Solver. The 15 alloying element palette upper bounds (Al + Mn, Cr, Zr, Mg, Si, Cu, Ti, Fe, Ni, Ce, Sc, Li, Er, Zn, Sn) define the outer feasibility envelope, baseline family priors and USGS-derived mineral burden. The live pKGFN BO loop in Phase I jointly optimizes the **full 15-element bounded composition vector and 20+ additional synthesis and processing variables across the powder-to-part pipeline** under a single asynchronous, cost-aware acquisition. Sparse Axis Aligned Subspace Bayesian Optimization (SAASBO), shown to work well in high dimensions [26, 35], will be used as the node-specific surrogate model for the downstream experimental nodes (highest fidelity, highest cost). A genetic algorithm will be used to solve the non-differentiable acquisition inner loop without resorting to pre-enumerating the candidate search space. The full compositional outer envelope implicitly spans $\sum_{k=2}^{8} \binom{15}{k} = 22\,803$ element combination subsystems (118,967,759 strictly supported compositions at 1 at.% resolution; $\sim 3.6 \times 10^{16}$ at 0.1 wt.%, Al $\geq$ 85 wt.%).

Pending evaluations are handled via fantasy samples (model predictions of pending points fed back into the model temporarily) so the acquisition stays asynchronous across simulation and experimental queues. The campaign will begin with several baselines followed by a set of 32 Sobol quasi-random designs to kickstart

the campaign with space-filling initial points before pKGFN takes over. TC-Python + TCAl10 Scheil + TC-PRISMA and the Thermo-Calc AM Module sit as three parallel cheap-to-costly contextual nodes to serve as inputs that inform the surrogate but are not themselves tunable decision variables. pKGFN chooses per candidate which to query under the cost budget. The use of `pycalphad` [36] with open-source databases will be explored; however, open-source Al databases are not as comprehensive as Thermo-Calc's TCAl10 and related databases. AM Module predictions map to Aconity MIDI parameters and are reciprocally calibrated by FLAIME prints. In line with the aims of Topic 3B, a mineral burden score is treated as a first-class BOFN objective [28, 35, 37]. Mineral lean family caps Mn ≤ 0.5, Cr ≤ 0.3, Zr ≤ 0.5, Sc ≤ 0.05 wt.% with gate G3b as the primary discovery constraint. Al–Mn–Cr–Zr [14] and Al–Zr serve as performance baseline comparators: both atomized early to seed the surrogate, ground the mineral burden benchmark, and validate the pipeline before committing the full campaign budget. A pre-registered pKGFN vs. standard BO ablation plan will be developed within the first two months of the project.

Activity 2: Powder → Print → Test Pipeline (*PI + partners; BYU ∼$280K, plus the Valimet subaward*). Powder feedstocks are produced via ultrasonic atomization using AMAZEMET rePOWDER, en-route from Poland as of 2026-04-24, for rapid composition iteration—mixtures of elemental powders are dosed at 0.1 mg resolution—providing high compositional design freedom. **Valimet (funded subrecipient) provides significant intellectual leadership** as co-developer of the gas-atomization scale-up envelope: joint definition of process windows for ∼6 production scale runs at ≥50 kg, joint specification of powder acceptance criteria (PSD, oxygen, ICP, satellites/morphology) and contribution to the manufacturability gates that feed back into the BOFN loop. The deliverable will be the 6 scale-up runs plus the additional shared design of the scale-up envelope and the specifications under which BYU ultrasonic-atomized screening compositions are deemed transferable to commercial gas atomization. Inductively coupled plasma mass spectrometry (ICP-MS) using BYU's Agilent 8900 provides authoritative bulk-composition verification against design target. Scanning Electron Microscopy combined with Energy Dispersive Spectroscopy (SEM/EDS) (Fig. 1) maps particle spatial distribution and is benchmarked against ICP-MS per the provenance audit. This will be done every k=10th composition with >5% relative EDS–ICP-MS disagreement triggering a re-run. Initial validation test mixtures composed of AlSi10Mg+Si+Al-wire have been atomized in advance through pre-collaboration with AMAZEMET. LPBF printing is performed at University of Utah's Multiscale Mechanics & Materials Lab led by Dr. Ashley Spear. Specimens follow a minimum-viable ladder (single tracks → thin-wall pillars → density cubes → ASTM E8/E8M sub-size tensile bars), with each stage gating the next on a partial evaluation of cost awareness and any stage early-stoppable. It creates a staged funnel: ∼160 atomized → ∼80 LPBF density cubes → ∼40 ASTM E8 tensile → top-3 elevated-temperature validation (Appendix 8 §A8.3). Sub-size ASTM E8/E8M (100 mm) specimens are standard for LPBF screening [38, 39]. Preliminary AlSi10Mg results validate the pipeline.

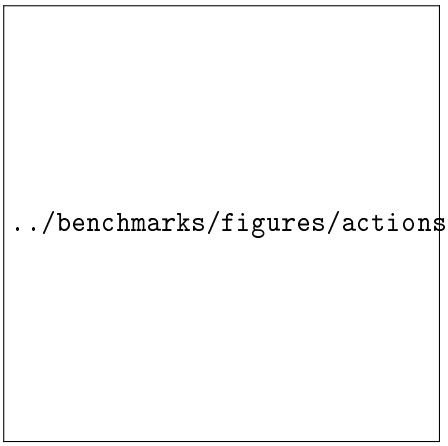


Figure 3: pKGFN vs. benchmarks: superior cost-efficiency via partial evaluations on a complex manufacturing problem [24], analogous to the AM alloy discovery pipeline proposed here.

Activity 3: Closed-Loop Integration, Dissemination & Qualification (*PI, BYU; ∼$100K*). The pKGFN engine orchestrates autonomous cycles: GP surrogates retrain after each batch and select next compositions and parameters and dispatched operating procedures map human handoffs 1:1 to future robotic points, minimizing technical debt. ISO 52902 benchmark geometries are printed as a BOFN node throughout the campaign, providing relevant parameter and qualification data stored with MongoDB. Data will be FAIR released to Zenodo/Figshare and AmSC-hosted Genesis Mission integration (App. 4 DMSP).

Project Management, Risk & Prior Work. Weekly PI–partner calls will be hosted. A FAIR-compliant database will be used as a single source of truth. Two Master students alongside about four full-time and four part-time undergraduate students in six-day/week shifts will form the lab staff to produce up to 16 compositions/week per Gantt, see Appendix 2. FLAIME builds on the PI's Honegumi SDL infrastructure and Fehrmann Materials ML-guided alloy contract but represents new closed-loop integration of composition discovery with AM qualification—no prior DOE funding supports this work.

4. Milestones in the Nine Months

Table 1: Phase I milestones and deliverables (9 months).

Mo.	ID	Milestone / Deliverable
1	M1	BOFN DAG deployed; pKGFN joint acquisition with GA inner solver live on BYU HPC; CALPHAD nodes (pycalphad, TCAl10, AM Module) queried across 15-element palette; first batch selected; Al–Mn–Cr–Zr and Al–Zr baseline-comparator atomization initiated
2	M2	$\geq$10 atomization runs (baselines + Sobol); powders characterized (PSD, SEM/EDS, ICP-MS); 5-baseline benchmark frozen; **powder-transfer validation:** Al–Mn–Cr–Zr "Alloy C" produced by both AMAZEMET ultrasonic and Valimet gas atomization compared (PSD D50 $\pm$15%, O $\pm$30%, density-window overlap $\geq$70%; if poor, branch to ultrasonic-only with gas-atomized Phase II handoff)
3	M3/G1	**Go/No-Go G1:** $\geq$50 compositions evaluated under joint pKGFN; GP surrogate RMSE $<$ pre-registered threshold (nested CV). *If fail: retrain with augmented CALPHAD; +1 mo.*
5	M4/G2	**Go/No-Go G2:** $\geq$1 end-to-end CALPHAD$\rightarrow$atomization$\rightarrow$LPBF$\rightarrow$tensile$\rightarrow$BOFN cycle; $\geq$2 compositions at $\geq$99.5% relative density. *If fail: decouple pipeline.*
7	M5/G3	**Go/No-Go G3:** $\geq$1 alloy meeting family thresholds (Al-Mn-Cr-Zr: YS$\geq$250 MPa, EL$\geq$5%; Al-Zr: YS$\geq$200 MPa, EL$\geq$8%) or AI-discovered equivalent; G3a MBS_{lin} $\leq$10% Scalmalloy *and* G3b $\text{CMDRI}_{\text{log}}$ $\leq$1.5$\times$ AlSi10Mg; ASTM E8 sub-size, $\geq$3 replicates. *If fail: pivot to strongest partial-pass.*
9	M6	Final deliverables: digital thread ($\geq$85%); dual-gate CMDRI validation; AI benchmark report (AF, win rate, bootstrap CI); FY27 continuation readiness

5. Data Sources and Models

Data sources. CALPHAD: Thermo-Calc TCAl10 (Al)/TCMG6 (Mg) via TC-Python under BYU's HPC license (Thermo-Calc letter, App. 7). DFT energies from Materials Project [40] and AFLOW [41] via Matminer [42]; experimental benchmarks: MMPDS, Knovel, ASM Alloy Center. Supply-risk scoring uses USGS MCS [1] and DOE Critical Materials Assessment [7]; BO integration follows [22]. These databases are used primarily to seed initial trials: literature-reported compositions are reproduced one-to-one as bias-free in-house ultrasonic-atomization replicates so all subsequent BOFN evaluations share a common feedstock-and-process baseline. FLAIME-generated data publish to Zenodo/Figshare with DOIs.

AI models. BOFN with pKGFN acquisition drives cost-aware selection atop GPyTorch/BoTorch GP surrogates [29], released open-source as Honegumi/BoTorch tutorials (Meta letter, App. 7). FLAIME hosts its digital thread on AmSC as a plug-and-play GM capability and leverages other GM teams' characterization data to accelerate surrogate training (App. 4 DMSP).

6. Decision Gate Metrics

Three pre-registered Phase I gates quantify AI advantage: **(1) AI Acceleration Factor**—$\text{AF}(y) = n_{\text{baseline}}/n_{\text{BOFN}}$ vs. four baselines (random, expert DOE, standard BO, BOFN ablation), $\geq$20 seeds with bootstrap CI [28]; *target* $\geq 3\times$. **(2) Discovery Yield**—fraction meeting family thresholds (Al-Mn-Cr-Zr: YS $\geq$ 250 MPa, EL $\geq$ 5%; ASTM E8 sub-size); *target* $\geq 3\%$. **(3) Mineral-Burden Reduction (dual G3)**—G3a $\text{MBS}_{\text{lin}} \leq 0.10\times$Scalmalloy *and* G3b $\text{CMDRI}_{\text{log}} \leq 1.5\times$AlSi10Mg (App. 8 §A8.1/§A8.4) [17]. GitHub Copilot & Edison used, human-reviewed.

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Appendix 2: Facilities & Other Resources

BYU Materials Discovery Laboratory (Lead Institution). The PI's dedicated laboratory space in the Department of Mechanical Engineering houses an **AMAZEMET rePOWDER ultrasonic atomizer with induction crucible module** (scheduled delivery April 30, 2026; week-long vendor-led commissioning and training scheduled). The rePOWDER accepts mixtures of elemental powders packed into machined high-purity aluminum crucibles (serving as both vial replacement and compaction chamber with press-fit lid), enabling high compositional design freedom; the **BYU Prototyping Lab** machines these crucibles in-house. The powder metallurgy lab is equipped for powder metallurgy and LPBF workflows—industrial dehumidification, fume hood, electrical, inert-gas, compressed-air, and chilled-water service installed in coordination with EH&S, sized for both the AMAZEMET rePOWDER induction-atomization workflow and the co-located LPBF workflow. Powder characterization is performed on-site via the **BYU Microscopy Center** (`https://microscopy.byu.edu`; SEM/EDS, EBSD, FIB); tensile and mechanical testing are directly adjacent (Instron 100 kip Landmark and 20 kip 809 A/T). Powder metallurgy and mechanical-testing spaces are connected by an internal door with no public-corridor traversal, a layout amenable to subsequent robotic integration for future hardware expansion.

Partner LPBF Access. Laser powder bed fusion is provided through **Prof. Ashley Spear's laboratory at the University of Utah** (Aconity MIDI; 170 mm ∅ build plate, open-parameter architecture, 200–1000 W fiber laser), ideal for rapid alloy screening; access and preliminary AlSi10Mg baseline prints are confirmed by letter (Appendix 7).

Computing Resources. BYU's **Office of Research Computing** operates a SLURM-managed HPC cluster (∼28,000 CPU cores, 367 GPUs incl. NVIDIA H200/H100/A100/L40S, >180 TB memory across 616 nodes; `https://rc.byu.edu`) allocated via fairshare at no direct project cost, supporting parallel TC-Python CALPHAD sweeps and BoTorch/GPyTorch surrogate training. If awarded, we will additionally request a **DOE HPC allocation** (e.g., NERSC or OLCF) to scale campaign-scale Bayesian optimization and AM-process simulations.

Partner Facilities (letters in Appendix 7). Valimet (Stockton, CA; gas-atomization scale-up of down-selected compositions, BYU subrecipient with ≥20% Valimet entity cost share); **Thermo-Calc Software/AB** (TCAl10, TCMG6, AM Module licenses and calibration collaboration; commercial vendor providing software and technical-staff support via BYU procurement); **AMAZEMET** (rePOWDER commissioning, operator training, and technical support; commercial vendor procured by BYU).

Safety & Powder Handling. Benchtop fume hoods, half-mask respirators with certified training, and a Nilfisk ATEX-rated metal-powder vacuum support safe handling; protocols were developed with BYU EH&S.

Appendix 3: Equipment

Major equipment by workflow (institution in parentheses):

- *Powder production:* **AMAZEMET rePOWDER** ultrasonic atomizer w/ induction-crucible module (BYU; delivery Apr 30, 2026; AMAZEMET engaged as commercial vendor / BYU procurement for commissioning + training); **Valimet** gas-atomization line (Valimet, Stockton CA; funded BYU subrecipient, scale-up).
- *LPBF/AM:* **Aconity MIDI** (UofU, Prof. Spear) — 170 mm ∅ plate, open-parameter, 200–1000 W fiber laser.
- *Characterization:* **BYU Microscopy Center** (SEM/EDS, EBSD, FIB, STEM; Verios G4, Helios NanoLab 600, Apreo C, Tecnai F20) — EDAX TEAM workflow already exercised on FLAIME feedstocks (Si-100, Si-40, AlSi10Mg); **BYU XRD Facility** (Chemistry); **Agilent 8900 ICP-MS/MS** (BYU).

- *Mechanical testing:* **Instron 100 kip Landmark** and **20 kip 809 A/T** servo-hydraulic testers (BYU).
- *Heat treatment:* **Recco** and **Lucifer** box furnaces (BYU).
- *Computing:* **BYU HPC** (∼28K cores, 367 GPUs incl. H200/H100/A100/L40S; SLURM fairshare).

Appendix 4: Data Management and Sharing Plan

1. Types of Data. FLAIME will generate: (i) alloy composition data (wt%/at% of Al, Mn, Cr, Zr via ICP-OES); (ii) CALPHAD simulation outputs (phase diagrams, Scheil–Gulliver curves, solidification parameters from Thermo-Calc TCAl10/AM Module); (iii) powder characterization records (PSD, SEM/EDS, flowability); (iv) LPBF processing parameters (power, speed, hatch spacing, layer thickness) and build metadata; (v) mechanical properties (YS, UTS, elongation per ASTM E8/E8M) and microstructural images (SEM, EBSD); and (vi) BOFN/pKGFN model parameters, acquisition function logs, and digital thread provenance records.

2. Data Formats. Tabular data will be stored in CSV/JSON with documented JSON schemas. Large simulation arrays will use HDF5. Microscopy images will be archived as lossless TIFF. ML models will be serialized in ONNX for interoperability.

3. Sharing and Preservation. All experimental datasets will be deposited on Zenodo and/or Figshare with DOIs within 12 months of collection. The FLAIME digital thread will be hosted on the American Science Cloud (AmSC) per Genesis Mission requirements. Software (BOFN/pKGFN engine, Honegumi tutorials) will be released on GitHub under MIT/Apache-2.0 licenses, following bssw.io best practices.

4. Access and Reuse. Data and models will adhere to FAIR principles. Datasets will carry rich README documentation and JSON schemas for machine discoverability. Data licensed CC-BY-4.0; code licensed MIT. Supply-risk metadata (Helbig SRI scores, $\mathrm{HHI}_{\mathrm{production}}$, $\mathrm{HHI}_{\mathrm{reserve}}$, and 3-year average metal cost derived from the frozen USGS MCS 2024 dataset) will be linked to every composition record.

5. Oversight. The PI is responsible for DMSP compliance. Quarterly data audits will be conducted throughout Phase I. Budget provisions for data curation, DOI registration, and AmSC hosting are included under Other Direct Costs.

Appendix 5: Synergistic Activities

Sterling Baird

1. **Self-Driving Laboratory Leadership & Community Building.** Serving as Guest Editor for *npj Computational Materials* for the "Self-Driving Labs for Chemistry and Materials Science" collection (2024–present) and lead author of the structural materials section of a comprehensive *Chemical Reviews* review on self-driving laboratories (Tom et al., 2024). Lead author of a review on low-cost self-driving laboratories introducing the "Frugal Twin" concept (Lo et al., 2024, *Digital Discovery*). Scientific Organizer for the Accelerate Conference (2023, 2024)—a major self-driving laboratory conference with hundreds of attendees. Solo-organized the Acceleration Consortium Bayesian Optimization Hackathon (120 participants, 62 organizations, 14 countries).
2. **Open-Source Software & Community Infrastructure.** Contributed over 17,000 open-source code artifacts (15K commits, 1.5K issues, 913 pull requests) spanning materials informatics, Bayesian optimization, and laboratory automation. Creator of `awesome-self-driving-labs`, a community-curated catalog of fully and partially autonomous laboratories. Developer of Honegumi, an interactive Bayesian optimization script generator for experimental scientists (arXiv:2502.06815, under review), and xtal2png, a crystal-structure-to-image encoding tool for generative modeling (JOSS, 2022). Served as Technical Consultant to Meta for the v1.0 release of the Ax/BoTorch platform (Nov 2024–Jan 2025). Over 350 self-driving lab demonstration kits deployed worldwide.
3. **Training & Workforce Development.** Delivered approximately 5,000 person-hours of self-driving laboratory training to roughly 700 individuals across 62 organizations through workshops, microcredential courses, and hackathons. Built the Acceleration Consortium Development Lab from the ground up ($600K in equipment, 500 line items, 24 months). Developed over 200 hours of course content in robotics, data science, and AI-driven experimental design.
4. **Standardization & Open Science.** Co-organizer of the annual NSF Pathways to Open-Source Ecosystems (POSE) workshop (2024, 2025, 2026). Contributing to Acceleration Consortium community standards for self-driving laboratory interoperability and advancing FAIR data practices for SDL datasets. Presenter at the German-Canadian Materials Acceleration Centre (GCMAC) Summer School (2024, 2025) and Review Panelist for the NSF TIP Directorate (2026).
5. **Open Data Impact & Domain Expertise.** Creator of the MPTS-52 time-based dataset split, widely adopted in the crystal structure generative AI community, and developer of the Matbench-genmetrics framework for standardized evaluation of generative models (JOSS, 2024). Completed a $34K industry research contract with Fehrmann Materials GmbH applying machine learning to joint strength and ductility prediction in aluminum alloys. Maintains curated open datasets on Zenodo and Figshare with documented community reuse.

D. Synergistic Activities